

Yuxin Xu

Email: yuxinxu@mail.dlut.edu.cn | Tel: (+86) 15546069298
Harbin, Heilongjiang Province, China

Education

Dalian University of Technology (Project 985), China

M.S. in Mechanical Engineering

- GPA: 3.79/4.00; Average Grade: 88.93/100; Rank: 34/343 (Top 10%)

Northeast Agricultural University (Project 211), China

B.S. in Mechanical Engineering

- GPA: 3.49/5.00; Average Grade: 84.99/100; Rank: 9/99 (Top 10%)

Dalian, Liaoning Province

Sep. 2024 - Jun. 2027 (expected)

Harbin, Heilongjiang Province

Sep. 2020 - Jun. 2024

Research Interests

Metastructure, Metamaterial, Biomechanics, Medical Device, Friction and Wear Analysis

Manuscripts in Preparation

- [1] Longfei Tan; **Yuxin Xu**; Bowen Hou; Changlin Du; Yupeng Li*, The hysteresis behavior analysis and friction coefficient reverse of bolted joint fitting surface with manufacturing errors. *Aerospace Science and Technology* (Submitted, 2026)
- [2] **Yuxin Xu**; Bowen Hou; Yuan Jin; Wei Zhang*, Substrate-Dependent Degradation and Lifetime Prediction of Silver Coatings on In718, TC4, and GCr15 Substrates under Vacuum Frictional Contact. (In Preparation)
- [3] **Yuxin Xu**; Longfei Tan; Bowen Hou; Jiahui Huang; Wei Zhang*, Effects of Manufacturing-Error-Based Thread Pitch Correction on the Hysteretic Behavior of Bolted Joints. (In Preparation)

* Corresponding Author

Research Experiences

Dalian University of Technology

Master's Research Advisor: Wei Zhang, Associate Professor, Associate Dean

Research Focus: Static/Dynamic Analysis and Friction/Wear Analysis of Bolted Joint Structures

Tribological Degradation Mechanisms and Lifetime Prediction of Silver Coatings Modulated by Substrate Materials

Leader

Apr. 2026 - Present

- Developed a residual-thickness-decay framework for silver-coating life prediction, incorporating substrate roughness, hardness, elastic modulus, and normal load to relate coating wear rate to metallic contact conditions.
- Defined failure life as the sliding distance required to reach a critical residual thickness and validated the framework using friction curves, residual-thickness measurements, and post-failure surface morphology, providing quantitative support for coating life design in metallic contacts.
- Outcome:** First-author manuscript in preparation.

Investigation into the Interfacial Degradation and Anomalous Disassembly Torque Increase of High-Temperature Dry-Film Coated Threaded Connections

Key Contributor

Jan. 2026 - Apr. 2026

- Developed a temperature-coating condition-interface wear-breakaway torque-bolt stress framework using asperity contact theory, adhesive-wear mechanisms, and the Tresca maximum shear stress criterion.
- Conducted four controlled bolt-self-locking-nut tests comparing intact and worn dry films at 400 °C and after further heating to 700 °C. The peak breakaway torque reached 4.62 times the room-temperature value; microscopy identified adhesive wear and material transfer in the crimped locking region, linking high-temperature coating degradation to the torque increase.
- Outcome:** A report has been completed.

Study on Nonlinear Hysteresis Behavior and Evolution of Bolted Joint Interface

Key Contributor

Sep. 2025 - Jan. 2026

- Built an ANSYS Workbench parametric finite-element model of a single-bolt lap joint that incorporated tightening, thread contact, bearing-surface nonparallelism, and roughness. Evaluated 16 combinations of tightening torque and cyclic tangential displacement to quantify hysteresis, equivalent stiffness, and loss factor.
- Quantified manufacturing-error effects: increasing bearing-surface nonparallelism from 0.5° to 1° increased the average change rate of equivalent stiffness from 11.28% to 19.32% and that of the loss factor from 9.86% to 12.32%, indicating accelerated evolution of joint parameters during service.
- Outcome:** First-author manuscript in preparation.

Development and Research on Advanced Threaded Connection Technology

Key Contributor

Apr. 2025 - Sep. 2025

- Developed theoretical and finite-element models of bolted-joint deformation and evaluated the effects of preload (315.92-568.65 kN), thread pitch (6.35-6.37 mm), structural parameters, and material properties on thread-load distribution, joint stiffness, strength, fatigue performance, and tightening behavior.
- Derived an equal-stress optimal pitch distribution for a structural-steel joint at 568.65 kN and validated it by finite-element analysis, reducing maximum thread stress by 13.83% and increasing thread stiffness by 8.9%. Torque-angle tightening reduced preload dispersion from 19.29% to 10.3% compared with torque control.

- **Outcome:** Four technical reports; findings were applied to the design of a wind-turbine blade-root bolting system and supported successful project acceptance.

Skills

Technical Skills

- **Programming Languages:** Python, MATLAB, C++, VB
- **Simulation & Design:** ANSYS, SolidWorks, Creo, CATIA, AutoCAD

Experimental Skills

- Friction and Wear Tester, Muffle Furnace, Electric Bolt Tightening Spindle, Fatigue Tensile Testing Machine, Scanning Electron Microscope (SEM), X-ray Fluorescence (XRF) Thickness Gauge

Language Skills

- Mandarin
- English (IELTS 6.5)

Awards and Scholarships

- National Second-Class Scholarship, Dalian University of Technology *Sep. 2025, Sep. 2026*
- Outstanding Student Award, Northeast Agricultural University *Sep. 2021, Sep. 2022, Sep. 2023*
- First Prize, The 16th National College Student Competition on Energy Conservation, Emission Reduction, Social Practice, and Technology, Northeast Agricultural University *Jun. 2023*
- Second Prize, The 10th National College Student Competition on Agricultural Building, Environment, and Energy Engineering, Northeast Agricultural University *Oct. 2022*